

Plant Biology

Photosynthesis, plant structures, transport, and reproduction

This material is designed as a concise student handout. It introduces core ideas, important vocabulary, and short review questions for classroom or self-study use.

Learning outcomes

- Explain the core concepts using accurate biology terminology.
- Connect biological structures to their functions.
- Apply the concepts to simple examples and review questions.
- Use the vocabulary section to strengthen scientific communication.

1. Plant structure and function

Plants have organs such as roots, stems, leaves, flowers, fruits, and seeds. Each part helps the plant survive, grow, and reproduce.

Roots anchor the plant and absorb water and minerals. Stems support the plant and transport materials. Leaves are major sites of photosynthesis.

Key points

- Roots absorb water and minerals.
- Stems support the plant and contain transport tissues.
- Leaves capture light and exchange gases.

Mini activity

Observe a leaf and sketch its shape, veins, and arrangement on the stem.

2. Photosynthesis

Photosynthesis is the process by which plants use light energy to make glucose from carbon dioxide and water. Oxygen is released as a by-product.

Chlorophyll in chloroplasts absorbs light. The glucose produced can be used for respiration, growth, storage, or building plant materials.

Key points

- Photosynthesis uses carbon dioxide, water, and light.
- Glucose stores chemical energy.
- Oxygen is released during photosynthesis.

Mini activity

Write the word equation for photosynthesis and identify the reactants and products.

3. Transport and reproduction

Xylem carries water and minerals from roots upward. Phloem transports sugars from leaves to other parts of the plant.

Many flowering plants reproduce sexually using flowers. Pollination transfers pollen to the female part of the flower, leading to fertilization and seed formation.

Key points

- Xylem transports water and minerals.
- Phloem transports sugars.
- Pollination is the transfer of pollen.
- Seeds contain the embryo of a new plant.

Mini activity

Label a flower diagram with petal, stamen, pollen, stigma, ovary, and ovule.

Important vocabulary

Term	Meaning
Chloroplast	Organelle where photosynthesis occurs.
Chlorophyll	Green pigment that absorbs light energy.
Xylem	Transport tissue for water and minerals.
Phloem	Transport tissue for sugars.
Pollination	Transfer of pollen to the female flower structure.

Quick review quiz

1. What are the main functions of roots, stems, and leaves?
2. What materials are needed for photosynthesis?
3. Why are chloroplasts important?
4. How are xylem and phloem different?
5. What is the role of pollination in plant reproduction?

Teacher note: Students should answer the review questions before checking a textbook, notes, or class discussion.